Title: PES-VCS Lab Report

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Screenshots

Phase 1

- Screenshot 1A – Output of ./test_objects showing all tests passed
- Screenshot 1B – Output of find .pes/objects -type f showing stored objects

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Phase 2

- Screenshot 2A – Output of ./test_tree showing all tests passed
- Screenshot 2B – Output of xxd on a tree object (first 20 lines)

Phase 3

- Screenshot 3A – Sequence of pes init, pes add file1.txt file2.txt, and pes status
- Screenshot 3B – Output of cat .pes/index showing index entries

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Phase 4

- Screenshot 4A – Output of ./pes log showing three commits
- Screenshot 4B – Output of find .pes -type f | sort showing repository contents

- Screenshot 4C – Output of cat .pes/refs/heads/main and cat .pes/HEAD

PHASE 1

A

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

B

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ S
```

PHASE 2

A

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

B

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ xxd .pes/objects/2a/594d39232
787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf |head -20
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 5879 0a                        egriXy.
```

PHASE 3

A

```
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  test_tree.c
  untracked:  pes.c
  untracked:  index.h
  untracked:  index.c
  untracked:  test_sequence.sh
  untracked:  test_objects.c
  untracked:  object.c
  untracked:  tree.h
  untracked:  README.md
  untracked:  tree.c
  untracked:  bye.txt
  untracked:  commit.c
  untracked:  Makefile
  untracked:  commit.h
  untracked:  hello.txt
  untracked:  pes.h

seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$
```

B

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776798445 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776798445 6 file2.txt
```

PHASE 4

A

```
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 50198e6c240b... Initial commit
Committed: 353e888d15a9... Add world
Committed: 780277fc4c92... Add farewell
commit 780277fc4c9267789104c3c6b55eb3f388090153f7c371c45e6fbe98a730b53d
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798277

    Add farewell

commit 353e888d15a9aafef84d53fd713a89dfaf48fadb5807f8d3a17f050d8a24b60d
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798277

    Add world

Add farewell

commit 50198e6c240bf0b53d2586d14f1d66d39c3bf2f942cb8b93f20b925e86cd2e53
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798277

    Initial commit farewell

commit 13da4821c4b465b314bf8e5fe58dd0a44cd0b8659152090ea77c2746449fdbb1
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798026

    first commitdd world

commit e6730aaeaed968de57ae21c9cde2a254d5896e77a6df58c1855785a02b6818e3
Author: Prarthana Shivakumar <PES1UG24CS339>
```

B

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/13/da4821c4b465b314bf8e5fe58dd0a44cd0b8659152090ea77c2746449fdbb1
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/35/3e888d15a9aafef84d53fd713a89dfaf48fadb5807f8d3a17f050d8a24b60d
.pes/objects/50/198e6c240bf0b53d2586d14f1d66d39c3bf2f942cb8b93f20b925e86cd2e53
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/78/0277fc4c9267789104c3c6b55eb3f388090153f7c371c45e6fbe98a730b53d
.pes/objects/dd/7493dd11a4e1855394b78721d8c44a16ff6db5efb484e209c638552d65d941
.pes/objects/e6/730aaeaed968de57ae21c9cde2a254d5896e77a6df58c1855785a02b6818e3
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

C

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
780277fc4c9267789104c3c6b55eb3f388090153f7c371c45e6fbe98a730b53d
ref: refs/heads/main
```

Final SS

```
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 75957784bb77... Initial commit

Log after first commit:
commit 75957784bb77582f9ab4f5dd6bcd162bc6efa38e0d9945b19937a8a139416ed0
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798530

    Initial commit


--- Second Commit ---
Committed: 367ffab83784... Update file.txt

--- Third Commit ---
Committed: 170953975acb... Add farewell
```

```
Committed: 170953975acb... Add farewell

--- Full History ---
commit 170953975acba5ca194f50cc84f48ae96a303ef1ad64aefbd5c8ff68c27d2450
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798530

    Add farewell

commit 367ffab83784080caaef3c8a962365f735d15c939f029321b91186fc4bd79b07
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798530

    Update file.txt

commit 75957784bb77582f9ab4f5dd6bcd162bc6efa38e0d9945b19937a8a139416ed0
Author: Prarthana Shivakumar <PES1UG24CS339>
Date:   1776798530

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
170953975acba5ca194f50cc84f48ae96a303ef1ad64aefbd5c8ff68c27d2450

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/17/0953975acba5ca194f50cc84f48ae96a303ef1ad64aefbd5c8ff68c27d2450
.pes/objects/36/7ffab83784080caaef3c8a962365f735d15c939f029321b91186fc4bd79b07
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/75/957784bb77582f9ab4f5dd6bcd162bc6efa38e0d9945b19937a8a139416ed0
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/dd/7493dd11a4e1855394b78721d8c44a16ff6db5efb484e209c638552d65d941

=== All integration tests completed ===
seed@seed-virtual-machine:~/PES1UG24CS339_pes_vcs$ cat .pes/index
```

Phase 5 — Branching

Q5.1
A branch is just a file storing a commit hash. To implement pes checkout <branch>, we update .pes/HEAD to point to the selected branch file. Then we read the commit it points to, get its tree, and update the working directory to match that snapshot. The difficult part is safely modifying files in the working directory without losing user changes and handling directories correctly.

Q5.2
A dirty working directory can be detected by comparing files in the working directory with the index. The index stores metadata like size, modification time, and hash. If a file has changed but is not staged, and checkout would overwrite it, then it is a conflict. In such cases, checkout should stop to prevent data loss.

Q5.3
Detached HEAD means HEAD points directly to a commit instead of a branch. Any new commits made are not linked to a branch and can be lost later. To recover them, a new branch can be created pointing to that commit as long as it hasn't been deleted.

Phase 6 — Garbage Collection

Q6.1
To remove unused objects, we start from all branch references and traverse all reachable commits, trees, and blobs, storing their hashes in a set. Then we scan the object store and delete any object not in this set. This ensures only unreachable objects are removed.

Q6.2
Running garbage collection during a commit can cause issues because some objects may be created but not yet referenced. GC might delete these valid objects, causing corruption. This race condition is avoided by using locking or delaying garbage collection until operations are complete.